Olivier Bisson

PhD Student in Geometric Statistics

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Profile

My research develops **differential-geometric tools for statistics on matrix manifolds**, with a focus on **symmetric positive-definite (SPD) covariance matrices** and **full-rank correlation matrices**. I study how geometric choices—especially **log-Euclidean metrics** and **quotient constructions**—shape geodesics and statistical summaries (e.g., Fréchet means and regression), and how these models behave near rank-deficient limits through **orbit-type stratification**. A recurring goal is to keep the geometry mathematically faithful while remaining computationally practical for applications such as longitudinal brain connectivity.

Research interests

Riemannian and quotient geometry on matrix manifolds; log-Euclidean metrics; correlation and covariance geometry; orbit-type stratification and rank-deficient limits; geometric statistics (means, regression, trajectories) with applications to medical imaging and neuroimaging.

Current position

Oct. 2023 – present — **PhD Student in Geometric Statistics**, *Inria (EPIONE) & Université Côte d'Azur*, France
Thesis: *Geometry, stratification and applications of structured correlation matrices.*
Advisor: Xavier Pennec.
Funding: 3IA (Interdisciplinary Institute for Artificial Intelligence).

Education

2023 — **M.Sc. in Mathematical Engineering**, *Stochastic and Statistical Modeling*, Université Côte d'Azur, France, Highest honors

2022 — **M.Sc. in Pure and Applied Mathematics**, *Algebra and Geometry*, Université Côte d'Azur, France

2020 — **B.Sc. in Fundamental Mathematics**, Université Côte d'Azur, France

(Pre)Publications

2025 — **O. Bisson**, X. Pennec. *Log-Euclidean Lie Groups*. arXiv preprint arXiv:2511.13380.

2025 — **O. Bisson**, Y. Aeschlimann, S. Deslauriers-Gauthier, X. Pennec. *Log-euclidean frameworks for smooth brain connectivity trajectories*. In: *Geometric Science of Information (GSI)*, St Malo, France, 2025, pp. 194–204. Springer. arXiv:2507.15374. *Nominated for Best Paper Award.*

2024 — **O. Bisson**, X. Pennec. *Differential of the Stretch Tensor for Any Dimension with Applications to Quotient Geodesics*. *Comptes Rendus. Mathématique*, 362 (2024), pp. 1847–1856. doi:10.5802/crmath.692. PDF.

Talks & conferences

Dec. 2025 — *Mini-course: Quotient Spaces and Related Differential Geometry*. EPIONE Seminar, Inria Sophia Antipolis, France.

Oct. 2025 — *Log-Euclidean Frameworks for Smooth Brain Connectivity Trajectories*. Geometric Science of Information (GSI), St Malo, France.

Mar. 2025 — *Geometry, Stratification and Applications of Structured Correlation Matrices*. EPIONE Seminar, Inria Sophia Antipolis, France.

Feb. 2025 — *Differentiable Structures on Correlation Matrices and Applications in Neuroimaging*. *Infinite-dimensional Geometry: Theory and Applications*, ESI (Vienna), Austria.

Dec. 2024 *Geometry, Stratification and Applications of Structured Correlation Matrices*. 3IA Côte d'Azur PhD Seminar, Nice, France.

Nov. 2024 *Riemannian Metrics on Correlation Matrices with Applications in Neuroimaging*. Mathematical Imaging and Random Geometry (RT MAIAGES), Nice, France.

Oct. 2024 *Introduction to the Riemannian Geometry of Correlation Matrices*. EPIONE Seminar, Inria Sophia Antipolis, France.

Feb. 2024 *A framework to put metrics on quotient spaces, with application in medical image analysis*. EPIONE Annual Retreat, Auron, France.

Dec. 2023 *Poster: Geometry, stratification and applications of structured correlation matrices*. 3IA Scientific Day, SKEMA, Sophia Antipolis, France.

Oct. 2023 *A framework to put metrics on quotient spaces, with application in medical image analysis*. EPIONE Seminar, Inria Sophia Antipolis, France.

Awards & Funding

2023–2026 **3IA Côte d'Azur PhD Funding**, Université Côte d'Azur / Inria.

2025 **Best Paper Award nomination** (GSI 2025) for *Log-euclidean frameworks for smooth brain connectivity trajectories*.

Teaching

2025–2026 **Teaching assistant (TD)**, *Introduction à l'analyse (L1)*, Université Côte d'Azur
Sets and functions; limits and continuity; differentiability; function study; injective/surjective/bijective maps; integrals and primitives.

2024–2025 **Teaching assistant (TD)**, *Compléments d'algèbre linéaire (L2)*, Université Côte d'Azur
Orthogonality and projections; Gram–Schmidt; least squares; isometries; introduction to PCA.

2024–2025 **Teaching assistant (TD)**, *Calculus II (L2)*, Université Côte d'Azur
Complex numbers; linear ODEs; multivariable calculus (partials, gradient, tangent planes, level sets, local extrema).

2023–2024 **Teaching assistant (TD)**, *Maths Approfondissement 2 (L1)*, Université Côte d'Azur
Exponential/logarithm refreshers; linear ODEs; Riemann integration basics; partial fractions; Euclidean vector geometry.

Software & open-source

Contributor to the `geomstats` Python package, with implementations related to Riemannian geometry on SPD and correlation matrix manifolds.

Previous Research experiences

Apr. 2023 – Oct. 2023 **Research intern**, *Inria (EPIONE), Université Côte d'Azur*, Nice/Valbonne, France
Topic: *Interpolation and extrapolation of correlation matrices for dynamic functional connectivity*.
Supervisors: Xavier Pennec, Samuel Deslauriers-Gauthier.
Implementation and evaluation of Riemannian metrics on correlation matrices for geodesic regression on resting-state fMRI correlation time-series.

Sep. 2022 – Feb. 2023 **Research project**, *LJAD & CEPAM, Université Côte d'Azur*, Nice, France
Topic: *Supervised classification of wood charcoal fragments: from machine to active learning*.
Supervisor: Marco Corneli.
Development of learning methods to automatically identify wood charcoal remains from a Southern-African context (M.Sc. capstone project).

Mar. 2022 – Jun. 2022 **Research intern**, *LJAD, Université Côte d'Azur*, Nice, France
Topic: *Quantum representations of mapping class groups*.
Supervisor: Jeremy Toulisse.
Topological/combinatorial approach (skein theory) to construct quantum representations of mapping class groups.

Languages

French	Native	*Fluent*
English	Fluent	*Reading, writing, speaking*
Italian	Basic	*Working knowledge*

Additional

Outside research: Volleyball, rock climbing, sailing, beach volleyball and swimming. I am deeply convinced that **anyone** can do mathematics.